Assignment
Probability

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Questions

1. An biased coin is tossed twice. The probability of getting heads is p and getting tails is q. The probability of getting one head and one tail is .

2. A box contains 5 red balls and 3 blue balls. A second box contains 2 red balls and 6 blue balls. If a box is chosen at random and a ball is drawn, and that ball is red, what is the probability it came from the first box? .

3. You are playing a game that involves rolling two fair six-sided dice. If the sum of the numbers on the dice is greater than or equal to 9, you win 10 rupees. Otherwise, you lose 5 rupees. What is the expected value of playing this game? Should you play this game?

4. The Cliff Walker Problem [Programming Question]:

   A person is standing at position $x = 1$ on a number line. At position $x = 0$, there is a cliff. At each step in time, the person moves one unit to the right (to $x + 1$) with probability p, or one unit to the left (to $x - 1$) with probability $q = 1 - p$. The game ends if they fall off the cliff.

   (a) If the walk is symmetric ($p = 1/2$), what is the probability that the person will eventually fall off the cliff?

   Instructions: Write a program to simulate this process. Since you cannot run an infinite game, for each simulated walk, set a maximum of 5,000 steps. If the person has not fallen off by then, you can assume they survived that particular walk. Run at least 100,000 simulations to get a stable estimate. Report the final estimated probability of the person falling off. Did you expect this result?

   (b) What value of p would make the probability of the person surviving (never falling off the cliff) approximately equal to $1/2$?

   Instructions: You may solve this by modifying your simulation from Part (a) to test different values of p.

   Note: The code might take a minute or two to run. Please be patient.

Bonus Questions [Optional]

1. Two players, A and B, are playing a coin toss game. A has $n + 1$ fair coins, and B has n fair coins. What is the probability that A will have more heads than B if both flip all their coins?

2. You are playing a game where you toss a fair six-sided die. You will get as many dollars as shown on the die, from 1 to 5 and the game ends. But if the die shows 6, you get 5 dollars and you get to roll the die again, this may happen indefinitely. What is the expected value of this game? Should you play this game if the entry fee is 5 dollars?

3. It is possible to solve the cliff walker problem analytically. Can you find the probability of falling off the cliff analytically if the $p = 2/3$ (Starting position is $x = 1$ and the cliff is at $x = 0$)? Also see if you can solve the problem for an arbitrary starting position $x = i$.

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